

I. Analyzing Reproductive Health and Behavior Patterns Using NSFG 2022–2023

1. Dataset Acquisition

```
library(tidyverse)
library(survey)
library(srvyr)
library(MASS)
library(nnet)
library(patchwork)
library(svyVGAM)

df_fem <- read_csv("/content/NSFG_2022_2023_FemRespPUFData.csv",
  guess_max = Inf)

Rows: 5586 Columns: 1912
— Column specification

```

```
Delimiter: ","
chr   (1): DEVICE_TYPE
dbl (1882): CaseID, RSCRAGE, RSCRNINF, RSCRHISP, RSCRRACE, FTFMODE,
AGE_R, A...
lgl   (29): BIONUMHX_5, MARENDHX_5, ENDMARRX_Y_5, SEDBCLC4, CMCURRP2,
CMCURR...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet
this message.
```

2. Variable Selection

- Count Variable
 - LIFPRTNR also known as the Lifetime number of opposite-sex sexual partners.
 - This is a non-negative integer that represents the frequency of a behavioral event over time.
 - It will support the Poisson Regression.
- Categorical (Nominal) Variable
 - CONSTAT1 also known as Current contraceptive status.
 - It has unordered categories and is used to analyze the differences in method choice across groups like marital status or race.
 - It will support Contingency Tables & Multinomial Logistic Regression.
- Ordered Categorical Variable
 - GENHEALT also known as Self-rated general health status.

- The responses follow the natural ranking system of 1-5 (Where 1=Excellent and 5=Poor)
 - It is required to use Ordinal Logistic Regression
 - Demographic Variables
 - AGER also known as Respondent's age at interview.
 - Age is a fundamental biological and social control variable, acting as a stable variable to determine factors affecting a group of people within certain age groups.
 - It will act as a Continuous Predictor.
-

- Demographic Variables
 - HISPRA2 also known as Race and Hispanic Origin
 - This variable categorizes the respondents by specific minority groups ensuring the data is applicable to these groups
 - It will support Multiple Comparisons and ANOVA
-

- Demographic Variables
 - HIEDUC also known as Highest completed year of school or degree
 - Educations acts as a way to gauge the socio-economic status of a respondent while also accounting for their healthcare literacy and access to healthcare services
 - It will as a Socio-Economic Predictor

3. Data Cleaning

```
df_clean <- df_fem %>%
  rename_with(toupper) %>%
  filter(MARSTAT <= 3) %>%
  mutate(
    contraception_group = case_when(
      CONSTAT1 %in% c(1, 2) ~ "Sterilization",
      CONSTAT1 %in% c(5, 6, 7, 10, 11, 12) ~ "Hormonal/LARC",
      CONSTAT1 == 3 ~ "Barrier",
      CONSTAT1 >= 13 ~ "Other/None",
      TRUE ~ NA_character_
    ),
    marital_fact = factor(MARSTAT, labels = c("Married", "Cohabiting",
      "Neither")),
    health_ord = factor(GENHEALT, levels = c(1, 2, 3, 4, 5), ordered =
      TRUE),
    race_fact = factor(HISPRA2, labels = c("Hispanic", "NH White",
      "NH Black", "Other"))
  ) %>%
  filter(!is.na(contraception_group), !is.na(GENHEALT), GENHEALT <= 5)

nsfg_design <- df_clean %>%
```

```
as_survey_design(
  ids = VECCL,
  strata = VEST,
  weights = WGT2022_2023,
  nest = TRUE
)
```

4. Statistical Analyses

A. Poisson Regression

```
mean_val <- svymean(~LIFPRTNR, nsfg_design, na.rm = TRUE)
var_val <- svyvar(~LIFPRTNR, nsfg_design, na.rm = TRUE)
```

```
print(mean_val)
print(var_val)
```

```
ratio <- as.numeric(var_val) / as.numeric(mean_val)
print(paste("Variance-to-Mean Ratio:", round(ratio, 2)))
```

```
      mean      SE
LIFPRTNR 5.969 0.1899
      variance      SE
LIFPRTNR   91.163 5.0394
[1] "Variance-to-Mean Ratio: 15.27"
```

```
pois_mod <- svyglm(LIFPRTNR ~ AGER + race_fact + HIEDUC, design =
  nsfg_design, family = poisson())
```

```
summary(pois_mod)
```

Call:

```
svyglm(formula = LIFPRTNR ~ AGER + race_fact + HIEDUC, design =
  nsfg_design,
  family = poisson())
```

Survey design:

Called via srvyr

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.240495	0.093840	2.563	0.013149	*
AGER	0.037910	0.002593	14.622	< 2e-16	***
race_factNH White	0.222246	0.072749	3.055	0.003468	**
race_factNH Black	0.333860	0.082102	4.066	0.000153	***
race_factOther	-0.159648	0.140052	-1.140	0.259258	
HIEDUC	0.019480	0.012165	1.601	0.115020	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 14.00669)

Number of Fisher Scoring iterations: 6

Goodness-of-fit:

14.007, indicating extreme overdispersion

This can also be seen in the assumption check which has the variance so much higher than the mean.

B. Contingency Tables

```
cont_table <- svytable(~MARSTAT + CONSTAT1, design = nsfg_design)
```

cont_table

CONSTAT1					
MARSTAT		1	2	3	5
6					
	1	4569426.886	3649447.764	275838.726	163135.177
		2557821.134			
	2	1383406.959	278352.345	295392.037	83882.141
		1080638.820			
	3	2531954.674	262689.352	1278762.816	705932.739
		4851958.143			
CONSTAT1					
MARSTAT		7	10	11	12
13					
	1	46744.291	2980505.462	35530.162	2859185.044
		0.000			
	2	17200.827	918512.117	0.000	724152.142
		8701.934			
	3	143863.934	2126509.766	0.000	1637538.925
		0.000			
CONSTAT1					
MARSTAT		19	20	21	22
30					
	1	230941.456	640443.663	1198299.924	29422.709
		1469763.070			
	2	0.000	203132.987	605964.701	0.000
		557092.923			
	3	69005.160	336506.819	743527.106	36023.042
		295284.362			
CONSTAT1					
MARSTAT		31	32	33	34
35					
	1	2139609.355	403292.398	1302722.932	194762.724
		460620.413			
	2	514061.628	85406.803	906288.865	74629.419
		164950.815			

3	467467.680	96650.825	1892848.483	0.000
226433.361				
CONSTAT1				
MARSTAT	38	40	41	42
1	18468.574	547156.757	1163412.388	2645837.613
2	0.000	398933.304	259324.490	968148.218
3	0.000	10878025.818	3258141.243	2448110.755

```
chisq_test <- svychisq(~marital_fact + contraception_group, design =
nsfg_design)
```

```
chisq_test
```

Pearson's X^2 : Rao & Scott adjustment

```
data: NextMethod()
```

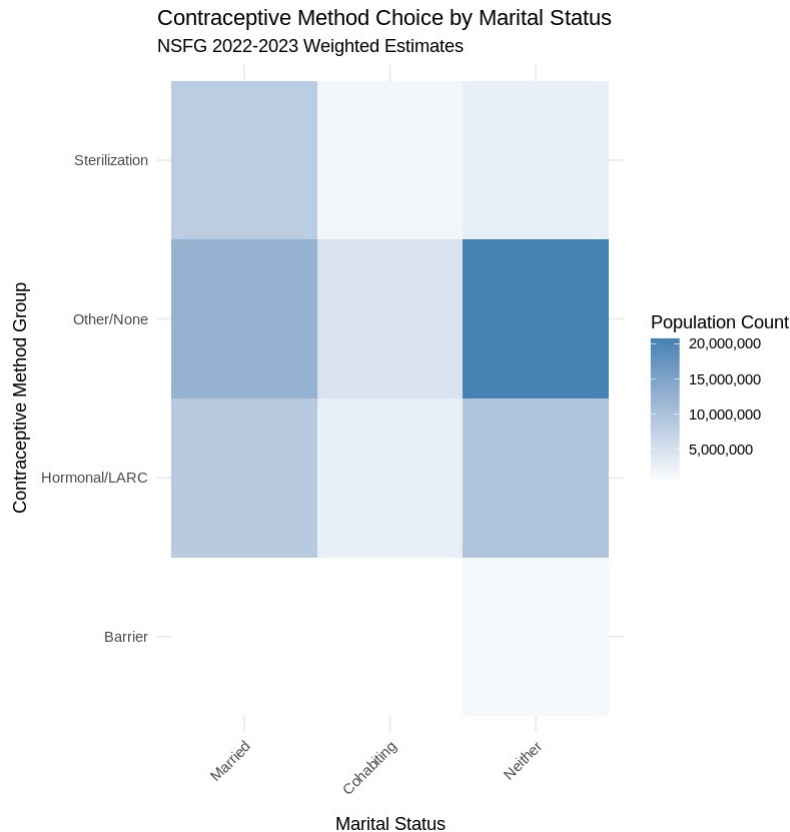
```
F = 43.354, ndf = 5.4075, ddf = 324.4529, p-value < 2.2e-16
```

Intepretation:

The p-value is less than 0.05 indicating that the marital status of a woman is not independent of her contraceptive method and is instead a good predictor of it.

```
cont_table_labeled <- svytable(~marital_fact + contraception_group,
design = nsfg_design)
```

```
as.data.frame(cont_table_labeled) %>%
  ggplot(aes(x = marital_fact, y = contraception_group, fill =
Freq)) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "steelblue", labels =
scales::comma) +
  labs(
    title = "Contraceptive Method Choice by Marital Status",
    subtitle = "NSFG 2022-2023 Weighted Estimates",
    x = "Marital Status",
    y = "Contraceptive Method Group",
    fill = "Population Count"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



C. Categorical Response Modeling

```
ord_mod <- svyolr(health_ord ~ AGER + HIEDUC + race_fact, design =
nsfg_design)
```

```
summary(ord_mod)
```

Call:

```
svyolr(health_ord ~ AGER + HIEDUC + race_fact, design = nsfg_design)
```

Coefficients:

	Value	Std. Error	t value
AGER	0.02913377	0.003723341	7.8246320
HIEDUC	-0.10610473	0.017872240	-5.9368457
race_factNH White	-0.05490063	0.114764548	-0.4783763
race_factNH Black	0.05771716	0.127858561	0.4514141
race_factOther	-0.03248479	0.135804204	-0.2392031

Intercepts:

	Value	Std. Error	t value
1 2	-1.0178	0.1499	-6.7884
2 3	0.6437	0.1461	4.4047
3 4	2.4161	0.1479	16.3336
4 5	4.3257	0.1745	24.7945

```
ord_summary <- cbind(
  "Log-Odds" = coef(ord_mod),
  "Odds Ratio" = exp(coef(ord_mod))
)
```

```
round(ord_summary, 3)
```

	Log-Odds	Odds Ratio
AGER	0.029	1.030
HIEDUC	-0.106	0.899
race_factNH White	-0.055	0.947
race_factNH Black	0.058	1.059
race_factOther	-0.032	0.968
1 2	-1.018	0.361
2 3	0.644	1.904
3 4	2.416	11.203
4 5	4.326	75.618

```
nom_mod <- svyVGAM::svy_vglm(contraception_group ~ AGER + HIEDUC +
  race_fact,
  design = nsfg_design,
  family = multinomial(refLevel = "Other/None"))
```

```
summary(nom_mod)
```

```
svy_vglm.survey.design(contraception_group ~ AGER + HIEDUC +
  race_fact, design = nsfg_design, family = multinomial(refLevel =
  "Other/None"))
```

Stratified 1 - level Cluster Sampling design (with replacement)

With (80) clusters.

Called via srvyr

Sampling variables:

- ids: VECL
- strata: VEST
- weights: WGT2022_2023

Data variables:

```
- CASEID (dbl), RSCRAGE (dbl), RSCRNINF (dbl), RSCRHISP (dbl),
RSCRACE
  (dbl), FTFMODE (dbl), DEVICE_TYPE (chr), AGE_R (dbl), AGESCRN
(dbl), HISP
  (dbl), HISPGRP (dbl), ROSCNT (dbl), NUMCHILD (dbl), HHKIDS18
(dbl),
  NONBIOKIDS (dbl), MARSTAT (dbl), LMARSTAT (dbl), RMARIT (dbl),
EVRMARRY
  (dbl), SSMARCOH (dbl), MANREL (dbl), EARNHS_Y (dbl), MYSCHOL_Y
(dbl),
  EARNBA_Y (dbl), WTHPARNW (dbl), ONOWN (dbl), ONOWN18 (dbl), INTACT
(dbl),
  PARMARR (dbl), INTACT18 (dbl), LVSIT14F (dbl), LVSIT14M (dbl),
WOMRASDU
```

(dbl), MOMWORKD (dbl), MANRASDU (dbl), FOSTEREV (dbl), MNYFSTER
 (dbl),
 DURFSTER (dbl), AGEFSTER (dbl), MARSTATB (dbl), EVCOHABB (dbl),
 FMARIT
 (dbl), MENARCHE (dbl), CURRPREG (dbl), LASTPREG (dbl), BIRTH1
 (dbl),
 OTHERKID (dbl), NOTHRKID (dbl), NBKIDLIV (dbl), NBKIDSEX_01 (dbl),
 NBKIDREL_01 (dbl), NBKDAGE_01 (dbl), ADPTNBKD_01 (dbl),
 TRYADOPT_01 (dbl),
 TRYEITHR_01 (dbl), NBKIDFOS_01 (dbl), NBKIDSEX_02 (dbl),
 NBKIDREL_02 (dbl),
 NBKDAGE_02 (dbl), ADPTNBKD_02 (dbl), TRYADOPT_02 (dbl),
 TRYEITHR_02 (dbl),
 NBKIDFOS_02 (dbl), NBKIDSEX_03 (dbl), NBKIDREL_03 (dbl),
 NBKDAGE_03 (dbl),
 ADPTNBKD_03 (dbl), TRYADOPT_03 (dbl), TRYEITHR_03 (dbl),
 NBKIDFOS_03 (dbl),
 EVERADOPT (dbl), BD_ADOPT (dbl), SEEKADPT (dbl), CONTAGEM (dbl),
 TRYLONG
 (dbl), KNOWADPT (dbl), EVWNTANO (dbl), EVCONTAG (dbl), TURNDOWN
 (dbl),
 YQUITTRY (dbl), HRDEMBRYO (dbl), SRCEMBRYO_1 (dbl), SRCEMBRYO_2
 (dbl),
 SRCEMBRYO_3 (dbl), SRCEMBRYO_4 (dbl), SRCEMBRYO_5 (dbl),
 FMARRDATE_Y (dbl),
 HERAGEFM (dbl), LIVTOGFS (dbl), STRTCFSP_Y (dbl), HERAGEF (dbl),
 CMSTRTFWP
 (dbl), ENGATFSP (dbl), WILLMARRF (dbl), CFSPEDUCN (dbl), CFSPMARBF
 (dbl),
 SSKIDTOG (dbl), NSSKIDTOG (dbl), SSKIDTOG18 (dbl), HSBVERIF (dbl),
 WHMARHX_Y_1 (dbl), HXAGEMAR_1 (dbl), LVTOGHX_1 (dbl), STRTOGHX_Y_1
 (dbl),
 ENGAGHX_1 (dbl), CHEDMARN_1 (dbl), MARBEFHX_1 (dbl), KIDSHX_1
 (dbl),
 KIDLIVHX_1 (dbl), BIOHUSBX_1 (dbl), BIONUMHX_1 (dbl), MARENDHX_1
 (dbl),
 ENDMARRX_Y_1 (dbl), WNSTPHX_Y_1 (dbl), WHMARHX_Y_2 (dbl),
 HXAGEMAR_2 (dbl),
 LVTOGHX_2 (dbl), STRTOGHX_Y_2 (dbl), ENGAGHX_2 (dbl), CHEDMARN_2
 (dbl),
 MARBEFHX_2 (dbl), KIDSHX_2 (dbl), KIDLIVHX_2 (dbl), BIOHUSBX_2
 (dbl),
 BIONUMHX_2 (dbl), MARENDHX_2 (dbl), ENDMARRX_Y_2 (dbl),
 WNSTPHX_Y_2 (dbl),
 WHMARHX_Y_3 (dbl), HXAGEMAR_3 (dbl), LVTOGHX_3 (dbl), STRTOGHX_Y_3
 (dbl),
 ENGAGHX_3 (dbl), CHEDMARN_3 (dbl), MARBEFHX_3 (dbl), KIDSHX_3
 (dbl),
 KIDLIVHX_3 (dbl), BIOHUSBX_3 (dbl), BIONUMHX_3 (dbl), MARENDHX_3

(dbl),
ENDMARRX_Y_3 (dbl), WNSTPHX_Y_3 (dbl), WHMARHX_Y_4 (dbl),
HXAGEMAR_4 (dbl),
LVTOGHX_4 (dbl), STRTOGHX_Y_4 (dbl), ENGAGHX_4 (dbl), CHEDMARN_4
(dbl),
MARBEFHX_4 (dbl), KIDSHX_4 (dbl), KIDLIVHX_4 (dbl), BIOHUSBX_4
(dbl),
BIONUMHX_4 (dbl), MARENDHX_4 (dbl), ENDMARRX_Y_4 (dbl),
WNSTPHX_Y_4 (dbl),
WHMARHX_Y_5 (dbl), HXAGEMAR_5 (dbl), LVTOGHX_5 (dbl), STRTOGHX_Y_5
(dbl),
ENGAGHX_5 (dbl), CHEDMARN_5 (dbl), MARBEFHX_5 (dbl), KIDSHX_5
(dbl),
KIDLIVHX_5 (dbl), BIOHUSBX_5 (dbl), BIONUMHX_5 (lgl), MARENDHX_5
(lgl),
ENDMARRX_Y_5 (lgl), WNSTPHX_Y_5 (dbl), CMMARRCH (dbl), PREVHUSB
(dbl),
WNSTRTCP_Y (dbl), CIPHERAGE (dbl), CPHISAGE (dbl), CPENGAG1 (dbl),
WILLMARR
(dbl), CPEDUC (dbl), CPMARBEF (dbl), CPKIDS (dbl), CPKIDLIV (dbl),
BIOCP
(dbl), BIONUMCP (dbl), LIVEOTH (dbl), EVRCOHAB (dbl), PREVCOHB
(dbl),
STRTOOTH1_Y (dbl), HERAGEC1 (dbl), HISAGEC1 (dbl), ENGAG1C1 (dbl),
MAREVC1
(dbl), C1KIDS (dbl), BIOFCP1 (dbl), BIONUMC1 (dbl), STPTOGC1_Y
(dbl),
EVERSEX (dbl), RHADSEX (dbl), YNOSEX (dbl), GRFSTSX (dbl),
TALKPAR1 (dbl),
TALKPAR2 (dbl), TALKPAR3 (dbl), TALKPAR4 (dbl), TALKPAR5 (dbl),
TALKPAR6
(dbl), TALKPAR7 (dbl), SEDNO (dbl), SEDNOLC1 (dbl), SEDNOLC2
(dbl),
SEDNOLC3 (dbl), SEDNOLC4 (dbl), SEDNOG (dbl), SEDNOSX (dbl), SEDBC
(dbl),
SEDBCLC1 (dbl), SEDBCLC2 (dbl), SEDBCLC3 (dbl), SEDBCLC4 (lgl),
SEDBCG
(dbl), SEDBCSX (dbl), SEDWHBC (dbl), SEDWHLC1 (dbl), SEDWHLC2
(dbl),
SEDWHLC3 (dbl), SEDWHLC4 (dbl), SEDWHBCG (dbl), SEDWBCSX (dbl),
SEDCOND
(dbl), SEDCONLC1 (dbl), SEDCONLC2 (dbl), SEDCONLC3 (dbl),
SEDCONLC4 (dbl),
SEDCONDG (dbl), SEDCONSX (dbl), SEDSTD (dbl), SEDSTDLC1 (dbl),
SEDSTDLC2
(dbl), SEDSTDLC3 (dbl), SEDSTDLC4 (dbl), SEDSTDG (dbl), SEDSTDSCX
(dbl),
SEDHIV (dbl), SEDHIVLC1 (dbl), SEDHIVLC2 (dbl), SEDHIVLC3 (dbl),
SEDHIVLC4

(dbl), SEDHIVG (dbl), SEDHIVSX (dbl), SEDABST (dbl), SEDABLC1
 (dbl),
 SEDABLC2 (dbl), SEDABLC3 (dbl), SEDABLC4 (dbl), SEDABSTG (dbl),
 SEDABSSX
 (dbl), SAMEMAN (dbl), WHOFTSTPR (dbl), KNOWFP (dbl), STILFPSX
 (dbl),
 LSTSEXP_M (dbl), LSTSEXP_Y (dbl), CMLSEXP (dbl), CMFPLAST
 (dbl),
 FP0THREL (dbl), FPEDUC (dbl), FPRN (dbl), WHICH1ST (dbl), SEXAFMEN
 (dbl),
 PTSB4MAR (dbl), WHOSNC1Y (dbl), MATCHFP (dbl), MATCHHP (dbl),
 P1YREL1
 (dbl), P1YLSEX_M (dbl), P1YLSEX_Y (dbl), CMLSEX1 (dbl), P1YCURRP1
 (dbl),
 PCURRNT1 (dbl), MATCHFP2 (dbl), MATCHHP2 (dbl), P1YREL2 (dbl),
 P1YLSEX_M2
 (dbl), P1YLSEX_Y2 (dbl), CMLSEX2 (dbl), P1YCURRP2 (dbl), PCURRNT2
 (dbl),
 MATCHFP3 (dbl), MATCHHP3 (dbl), P1YREL3 (dbl), P1YLSEX_M3 (dbl),
 P1YLSEX_Y3 (dbl), CMLSEX3 (dbl), P1YCURRP3 (dbl), PCURRNT3 (dbl),
 P1Y0THREL
 (dbl), P1YRAGE (dbl), P1YHSAGE (dbl), P1YRF (dbl), P1YFSEX_M
 (dbl),
 P1YFSEX_Y (dbl), CMFSEX (dbl), CMFSEXTOT (dbl), P1YEDUC (dbl),
 P1YRN1
 (dbl), P1Y0THREL2 (dbl), P1YRAGE2 (dbl), P1YHSAGE2 (dbl), P1YRF2
 (dbl),
 P1YFSEX_M2 (dbl), P1YFSEX_Y2 (dbl), CMFSEX2 (dbl), CMFSEXTOT2
 (dbl),
 P1YEDUC2 (dbl), P1YRN2 (dbl), P1Y0THREL3 (dbl), P1YRAGE3 (dbl),
 P1YHSAGE3
 (dbl), P1YRF3 (dbl), P1YFSEX_M3 (dbl), P1YFSEX_Y3 (dbl), CMFSEX3
 (dbl),
 CMFSEXTOT3 (dbl), P1YEDUC3 (dbl), P1YRN3 (dbl), CURRPRTT (dbl),
 CURRPRTS
 (dbl), CMCURRP1 (dbl), CMCURRP2 (lg), CMCURRP3 (lg), EVERTUBS
 (dbl),
 TUBSDATE_Y (dbl), EVERHYST (dbl), HYSTDAT_Y (dbl), HYST (dbl),
 SAMEOPER
 (dbl), OTHROPS1_1 (dbl), OTHROPS1_2 (dbl), OTHROPS2_1 (dbl),
 OTHROPS2_2
 (dbl), OTHSTER (dbl), RHADALL (dbl), HHADALL (dbl), FMEDREAS
 (dbl), BCREAS
 (dbl), BCWHYF (dbl), TUBLMAIN (dbl), PAYTUBAL_1 (dbl), PAYTUBAL_2
 (dbl),
 ANYFSTER (dbl), ANYVASEC (dbl), VASDATE_Y (dbl), WITHIMOP (dbl),
 VASJAN4YR
 (lg), PAYVAS_1 (dbl), PAYVAS_2 (dbl), REVSTUBL (dbl), DATRVSTB_Y
 (dbl),

RESVASX (dbl), DATRVVEX_Y (dbl), TUBS (dbl), VASECT (dbl),
RSURGSTR (dbl),
PSURGSTR (dbl), ONLYTBVS (dbl), RWANTRVT (dbl), MANWANTT (dbl),
RWANTREV
(dbl), MANWANTR (dbl), POSIBLPG (dbl), REASIMPR (dbl), POSIBLMN
(dbl),
REASIMPP (dbl), CANHAVER (dbl), REASDIFF_1 (dbl), REASDIFF_2
(dbl),
REASDIFF_3 (dbl), REASDIFF_4 (dbl), REASDIFF_5 (dbl), CANHAVEM
(dbl),
PREGNONO (dbl), REASNONO_1 (dbl), REASNONO_2 (dbl), REASNONO_3
(dbl),
RSTRSTAT (dbl), PSTRSTAT (dbl), LASTPER (dbl), TRYPREG12 (dbl),
PILL (dbl),
CONDOM (dbl), VASECTMY (dbl), DEPOPROV (dbl), WITHDRAWAL (dbl),
RHYTHM (dbl),
SDAYCBDS (dbl), TEMPSAFE (dbl), NATCYCA (dbl), PATCH (dbl), RING
(dbl),
MORNPIII (dbl), ECTIMESX (dbl), ECRX (dbl), ECWHERE (dbl), ECWHEN
(dbl),
IMPLANT (dbl), IUD (dbl), EVIUDTYP1 (dbl), EVIUDTYP2 (dbl),
EVCIUDEC (dbl),
OTHRMETH01 (dbl), OTHRMETH02 (dbl), OTHRMETH03 (dbl), OTHRMETH04
(dbl),
OTHRMETH05 (dbl), OTHRMETH06 (dbl), EVERUSED (dbl), METHDISS
(dbl),
METHSTOP01 (dbl), METHSTOP02 (dbl), METHSTOP03 (dbl), METHSTOP04
(dbl),
METHSTOP05 (dbl), METHSTOP06 (dbl), METHSTOP07 (dbl), METHSTOP08
(dbl),
METHSTOP09 (dbl), WHENPILL (dbl), REASPIII01 (dbl), REASPIII02
(dbl),
REASPIII03 (dbl), REASPIII04 (dbl), REASPIII05 (dbl), REASPIII06
(dbl),
REASPIII07 (dbl), WHENCOND (dbl), REASCOND01 (dbl), REASCOND02
(dbl),
REASCOND03 (dbl), REASCOND04 (dbl), REASCOND05 (dbl), REASCOND06
(dbl),
REASCOND07 (dbl), REASCOND08 (dbl), WHENIUD (dbl), TYPEIUD_1
(dbl),
TYPEIUD_2 (dbl), REASIUD01 (dbl), REASIUD02 (dbl), REASIUD03
(dbl),
REASIUD04 (dbl), REASIUD05 (dbl), REASIUD06 (dbl), FIRSMETH1
(dbl),
FIRSMETH2 (dbl), FIRSMETH3 (dbl), FIRSMETH4 (dbl), NUMFIRSM (dbl),
DRUGDEV
(dbl), DRUGDEV2 (dbl), DRUGDEV3 (dbl), DRUGDEV4 (dbl), FIRSTIME1
(dbl),
FIRSTIME2 (dbl), WNFSTUSE_M (dbl), WNFSTUSE_Y (dbl), CMFIRSM

(dbl),
CMFSTUSE (dbl), AGEFSTUS (dbl), USEFRSTS (dbl), USEFSTSX (dbl),
MTHFRSTS1
(dbl), MTHFRSTS2 (dbl), MTHFRSTS3 (dbl), MTHFRSTS4 (dbl),
MTHFSTSX1 (dbl),
MTHFSTSX2 (dbl), MTHFSTSX3 (dbl), MTHFSTSX4 (dbl), INTR_EC3 (dbl),
CMMONSX
(dbl), MONSX (dbl), CMMONSX2 (dbl), MONSX2 (dbl), CMMONSX3 (dbl),
MONSX3
(dbl), CMMONSX4 (dbl), MONSX4 (dbl), CMMONSX5 (dbl), MONSX5 (dbl),
CMMONSX6
(dbl), MONSX6 (dbl), CMMONSX7 (dbl), MONSX7 (dbl), CMMONSX8 (dbl),
MONSX8
(dbl), CMMONSX9 (dbl), MONSX9 (dbl), CMMONSX10 (dbl), MONSX10
(dbl),
CMMONSX11 (dbl), MONSX11 (dbl), CMMONSX12 (dbl), MONSX12 (dbl),
CMMONSX13
(dbl), MONSX13 (dbl), CMMONSX14 (dbl), MONSX14 (dbl), CMMONSX15
(dbl),
MONSX15 (dbl), CMMONSX16 (dbl), MONSX16 (dbl), CMMONSX17 (dbl),
MONSX17
(dbl), CMMONSX18 (dbl), MONSX18 (dbl), CMMONSX19 (dbl), MONSX19
(dbl),
CMMONSX20 (dbl), MONSX20 (dbl), CMMONSX21 (dbl), MONSX21 (dbl),
CMMONSX22
(dbl), MONSX22 (dbl), CMMONSX23 (dbl), MONSX23 (dbl), CMMONSX24
(dbl),
MONSX24 (dbl), CMMONSX25 (dbl), MONSX25 (dbl), CMMONSX26 (dbl),
MONSX26
(dbl), CMMONSX27 (dbl), MONSX27 (dbl), CMMONSX28 (dbl), MONSX28
(dbl),
CMMONSX29 (dbl), MONSX29 (dbl), CMMONSX30 (dbl), MONSX30 (dbl),
CMMONSX31
(dbl), MONSX31 (dbl), CMMONSX32 (dbl), MONSX32 (dbl), CMMONSX33
(dbl),
MONSX33 (dbl), CMMONSX34 (dbl), MONSX34 (dbl), CMMONSX35 (dbl),
MONSX35
(dbl), CMMONSX36 (dbl), MONSX36 (dbl), CMMONSX37 (dbl), MONSX37
(dbl),
CMMONSX38 (dbl), MONSX38 (dbl), CMMONSX39 (dbl), MONSX39 (dbl),
CMMONSX40
(dbl), MONSX40 (dbl), CMMONSX41 (dbl), MONSX41 (dbl), CMMONSX42
(dbl),
MONSX42 (dbl), CMMONSX43 (dbl), MONSX43 (dbl), CMMONSX44 (dbl),
MONSX44
(dbl), CMMONSX45 (dbl), MONSX45 (dbl), CMMONSX46 (dbl), MONSX46
(dbl),
CMMONSX47 (dbl), MONSX47 (dbl), CMMONSX48 (dbl), MONSX48 (dbl),
CMSTRTMC

(dbl), CMENDMC (dbl), INTR_ED4A (dbl), NUMMCMON (dbl), MC1MONS1 (dbl), MC1SIMSQ (dbl), MC1MONS2 (dbl), MC1MONS3 (dbl), DATBEGIN_M (dbl), DATBEGIN_Y (dbl), CMDATBEGIN (dbl), CURRMETH1 (dbl), CURRMETH2 (dbl), CURRMETH3 (dbl), CURRMETH4 (dbl), LSTMONMETH1 (dbl), LSTMONMETH2 (dbl), LSTMONMETH3 (dbl), LSTMONMETH4 (dbl), PILL_12 (dbl), DIAPH_12 (dbl), IUD_12 (dbl), IMPLANT_12 (dbl), DEPO_12 (dbl), CERVC_12 (dbl), MPILL_12 (dbl), PATCH_12 (dbl), RING_12 (dbl), METHX1 (dbl), METHX2 (dbl), METHX3 (dbl), METHX4 (dbl), METHX5 (dbl), METHX6 (dbl), METHX7 (dbl), METHX8 (dbl), METHX9 (dbl), METHX10 (dbl), METHX11 (dbl), METHX12 (dbl), METHX13 (dbl), METHX14 (dbl), METHX15 (dbl), METHX16 (dbl), METHX17 (dbl), METHX18 (dbl), METHX19 (dbl), METHX20 (dbl), METHX21 (dbl), METHX22 (dbl), METHX23 (dbl), METHX24 (dbl), METHX25 (dbl), METHX26 (dbl), METHX27 (dbl), METHX28 (dbl), METHX29 (dbl), METHX30 (dbl), METHX31 (dbl), METHX32 (dbl), METHX33 (dbl), METHX34 (dbl), METHX35 (dbl), METHX36 (dbl), METHX37 (dbl), METHX38 (dbl), METHX39 (dbl), METHX40 (dbl), METHX41 (dbl), METHX42 (dbl), METHX43 (dbl), METHX44 (dbl), METHX45 (dbl), METHX46 (dbl), METHX47 (dbl), METHX48 (dbl), METHX49 (dbl), METHX50 (dbl), METHX51 (dbl), METHX52 (dbl), METHX53 (dbl), METHX54 (dbl), METHX55 (dbl), METHX56 (dbl), METHX57 (dbl), METHX58 (dbl), METHX59 (dbl), METHX60 (dbl), METHX61 (dbl), METHX62 (dbl), METHX63 (dbl), METHX64 (dbl), METHX65 (dbl), METHX66 (dbl), METHX67 (dbl), METHX68 (dbl), METHX69 (dbl), METHX70 (dbl), METHX71 (dbl), METHX72 (dbl), METHX73 (dbl), METHX74 (dbl), METHX75 (dbl), METHX76 (dbl), METHX77 (dbl), METHX78 (dbl), METHX79 (dbl), METHX80 (dbl), METHX81 (dbl), METHX82 (dbl), METHX83 (dbl), METHX84 (dbl), METHX85 (dbl), METHX86 (dbl), METHX87 (dbl), METHX88 (dbl), METHX89 (dbl), METHX90 (dbl), METHX91 (dbl), METHX92 (dbl), METHX93 (dbl),

METHX94 (dbl), METHX95 (dbl), METHX96 (dbl), METHX97 (dbl),
METHX98 (dbl),
METHX99 (dbl), METHX100 (dbl), METHX101 (dbl), METHX102 (dbl),
METHX103
(dbl), METHX104 (dbl), METHX105 (dbl), METHX106 (dbl), METHX107
(dbl),
METHX108 (dbl), METHX109 (dbl), METHX110 (dbl), METHX111 (dbl),
METHX112
(dbl), METHX113 (dbl), METHX114 (dbl), METHX115 (dbl), METHX116
(dbl),
METHX117 (dbl), METHX118 (dbl), METHX119 (dbl), METHX120 (dbl),
METHX121
(dbl), METHX122 (dbl), METHX123 (dbl), METHX124 (dbl), METHX125
(dbl),
METHX126 (dbl), METHX127 (dbl), METHX128 (dbl), METHX129 (dbl),
METHX130
(dbl), METHX131 (dbl), METHX132 (dbl), METHX133 (dbl), METHX134
(dbl),
METHX135 (dbl), METHX136 (dbl), METHX137 (dbl), METHX138 (dbl),
METHX139
(dbl), METHX140 (dbl), METHX141 (dbl), METHX142 (dbl), METHX143
(dbl),
METHX144 (dbl), METHX145 (dbl), METHX146 (dbl), METHX147 (dbl),
METHX148
(dbl), METHX149 (dbl), METHX150 (dbl), METHX151 (dbl), METHX152
(dbl),
METHX153 (dbl), METHX154 (dbl), METHX155 (dbl), METHX156 (dbl),
METHX157
(dbl), METHX158 (dbl), METHX159 (dbl), METHX160 (dbl), METHX161
(dbl),
METHX162 (dbl), METHX163 (dbl), METHX164 (dbl), METHX165 (dbl),
METHX166
(dbl), METHX167 (dbl), METHX168 (dbl), METHX169 (dbl), METHX170
(dbl),
METHX171 (dbl), METHX172 (dbl), METHX173 (dbl), METHX174 (dbl),
METHX175
(dbl), METHX176 (dbl), METHX177 (dbl), METHX178 (dbl), METHX179
(dbl),
METHX180 (dbl), METHX181 (dbl), METHX182 (dbl), METHX183 (dbl),
METHX184
(dbl), METHX185 (dbl), METHX186 (dbl), METHX187 (dbl), METHX188
(dbl),
METHX189 (dbl), METHX190 (dbl), METHX191 (dbl), METHX192 (dbl),
CMMHCALX1
(dbl), CMMHCALX2 (dbl), CMMHCALX3 (dbl), CMMHCALX4 (dbl),
CMMHCALX5 (dbl),
CMMHCALX6 (dbl), CMMHCALX7 (dbl), CMMHCALX8 (dbl), CMMHCALX9
(dbl),
CMMHCALX10 (dbl), CMMHCALX11 (dbl), CMMHCALX12 (dbl), CMMHCALX13

(dbl),
CMMHCALX14 (dbl), CMMHCALX15 (dbl), CMMHCALX16 (dbl), CMMHCALX17
(dbl),
CMMHCALX18 (dbl), CMMHCALX19 (dbl), CMMHCALX20 (dbl), CMMHCALX21
(dbl),
CMMHCALX22 (dbl), CMMHCALX23 (dbl), CMMHCALX24 (dbl), CMMHCALX25
(dbl),
CMMHCALX26 (dbl), CMMHCALX27 (dbl), CMMHCALX28 (dbl), CMMHCALX29
(dbl),
CMMHCALX30 (dbl), CMMHCALX31 (dbl), CMMHCALX32 (dbl), CMMHCALX33
(dbl),
CMMHCALX34 (dbl), CMMHCALX35 (dbl), CMMHCALX36 (dbl), CMMHCALX37
(dbl),
CMMHCALX38 (dbl), CMMHCALX39 (dbl), CMMHCALX40 (dbl), CMMHCALX41
(dbl),
CMMHCALX42 (dbl), CMMHCALX43 (dbl), CMMHCALX44 (dbl), CMMHCALX45
(dbl),
CMMHCALX46 (dbl), CMMHCALX47 (dbl), CMMHCALX48 (dbl), NUMMULTX1
(dbl),
NUMMULTX2 (dbl), NUMMULTX3 (dbl), NUMMULTX4 (dbl), NUMMULTX5
(dbl),
NUMMULTX6 (dbl), NUMMULTX7 (dbl), NUMMULTX8 (dbl), NUMMULTX9
(dbl),
NUMMULTX10 (dbl), NUMMULTX11 (dbl), NUMMULTX12 (dbl), NUMMULTX13
(dbl),
NUMMULTX14 (dbl), NUMMULTX15 (dbl), NUMMULTX16 (dbl), NUMMULTX17
(dbl),
NUMMULTX18 (dbl), NUMMULTX19 (dbl), NUMMULTX20 (dbl), NUMMULTX21
(dbl),
NUMMULTX22 (dbl), NUMMULTX23 (dbl), NUMMULTX24 (dbl), NUMMULTX25
(dbl),
NUMMULTX26 (dbl), NUMMULTX27 (dbl), NUMMULTX28 (dbl), NUMMULTX29
(dbl),
NUMMULTX30 (dbl), NUMMULTX31 (dbl), NUMMULTX32 (dbl), NUMMULTX33
(dbl),
NUMMULTX34 (dbl), NUMMULTX35 (dbl), NUMMULTX36 (dbl), NUMMULTX37
(dbl),
NUMMULTX38 (dbl), NUMMULTX39 (dbl), NUMMULTX40 (dbl), NUMMULTX41
(dbl),
NUMMULTX42 (dbl), NUMMULTX43 (dbl), NUMMULTX44 (dbl), NUMMULTX45
(dbl),
NUMMULTX46 (dbl), NUMMULTX47 (dbl), NUMMULTX48 (dbl), SALMX1
(dbl), SALMX2
(dbl), SALMX3 (dbl), SALMX4 (dbl), SALMX5 (dbl), SALMX6 (dbl),
SALMX7
(dbl), SALMX8 (dbl), SALMX9 (dbl), SALMX10 (dbl), SALMX11 (dbl),
SALMX12
(dbl), SALMX13 (dbl), SALMX14 (dbl), SALMX15 (dbl), SALMX16 (dbl),
SALMX17

(dbl), SALMX18 (dbl), SALMX19 (dbl), SALMX20 (dbl), SALMX21 (dbl),
SALMX22
(dbl), SALMX23 (dbl), SALMX24 (dbl), SALMX25 (dbl), SALMX26 (dbl),
SALMX27
(dbl), SALMX28 (dbl), SALMX29 (dbl), SALMX30 (dbl), SALMX31 (dbl),
SALMX32
(dbl), SALMX33 (dbl), SALMX34 (dbl), SALMX35 (dbl), SALMX36 (dbl),
SALMX37
(dbl), SALMX38 (dbl), SALMX39 (dbl), SALMX40 (dbl), SALMX41 (dbl),
SALMX42
(dbl), SALMX43 (dbl), SALMX44 (dbl), SALMX45 (dbl), SALMX46 (dbl),
SALMX47
(dbl), SALMX48 (dbl), SAYX1 (dbl), SAYX2 (dbl), SAYX3 (dbl), SAYX4
(dbl),
SAYX5 (dbl), SAYX6 (dbl), SAYX7 (dbl), SAYX8 (dbl), SAYX9 (dbl),
SAYX10
(dbl), SAYX11 (dbl), SAYX12 (dbl), SAYX13 (dbl), SAYX14 (dbl),
SAYX15
(dbl), SAYX16 (dbl), SAYX17 (dbl), SAYX18 (dbl), SAYX19 (dbl),
SAYX20
(dbl), SAYX21 (dbl), SAYX22 (dbl), SAYX23 (dbl), SAYX24 (dbl),
SAYX25
(dbl), SAYX26 (dbl), SAYX27 (dbl), SAYX28 (dbl), SAYX29 (dbl),
SAYX30
(dbl), SAYX31 (dbl), SAYX32 (dbl), SAYX33 (dbl), SAYX34 (dbl),
SAYX35
(dbl), SAYX36 (dbl), SAYX37 (dbl), SAYX38 (dbl), SAYX39 (dbl),
SAYX40
(dbl), SAYX41 (dbl), SAYX42 (dbl), SAYX43 (dbl), SAYX44 (dbl),
SAYX45
(dbl), SAYX46 (dbl), SAYX47 (dbl), SAYX48 (dbl), SIMSEQX1 (dbl),
SIMSEQX2
(dbl), SIMSEQX3 (dbl), SIMSEQX4 (dbl), SIMSEQX5 (dbl), SIMSEQX6
(dbl),
SIMSEQX7 (dbl), SIMSEQX8 (dbl), SIMSEQX9 (dbl), SIMSEQX10 (dbl),
SIMSEQX11
(dbl), SIMSEQX12 (dbl), SIMSEQX13 (dbl), SIMSEQX14 (dbl),
SIMSEQX15 (dbl),
SIMSEQX16 (dbl), SIMSEQX17 (dbl), SIMSEQX18 (dbl), SIMSEQX19
(dbl),
SIMSEQX20 (dbl), SIMSEQX21 (dbl), SIMSEQX22 (dbl), SIMSEQX23
(dbl),
SIMSEQX24 (dbl), SIMSEQX25 (dbl), SIMSEQX26 (dbl), SIMSEQX27
(dbl),
SIMSEQX28 (dbl), SIMSEQX29 (dbl), SIMSEQX30 (dbl), SIMSEQX31
(dbl),
SIMSEQX32 (dbl), SIMSEQX33 (dbl), SIMSEQX34 (dbl), SIMSEQX35
(dbl),
SIMSEQX36 (dbl), SIMSEQX37 (dbl), SIMSEQX38 (dbl), SIMSEQX39

(dbl),
SIMSEQX40 (dbl), SIMSEQX41 (dbl), SIMSEQX42 (dbl), SIMSEQX43
(dbl),
SIMSEQX44 (dbl), SIMSEQX45 (dbl), SIMSEQX46 (dbl), SIMSEQX47
(dbl),
SIMSEQX48 (dbl), USELSTP (dbl), WYNOLSTP (dbl), HPLSTP (dbl),
LSTMTHP1
(dbl), LSTMTHP2 (dbl), LSTMTHP3 (dbl), LSTMTHP4 (dbl), USEFSTP
(dbl),
FSTMTHP1 (dbl), FSTMTHP2 (dbl), FSTMTHP3 (dbl), FSTMTHP4 (dbl),
USELSTP2
(dbl), WYNOLSTP2 (dbl), HPLSTP2 (dbl), LSTMTHP5 (dbl), LSTMTHP6
(dbl),
LSTMTHP7 (dbl), LSTMTHP8 (dbl), USEFSTP2 (dbl), FSTMTHP5 (dbl),
FSTMTHP6
(dbl), FSTMTHP7 (dbl), FSTMTHP8 (dbl), USELSTP3 (dbl), WYNOLSTP3
(dbl),
HPLSTP3 (dbl), LSTMTHP9 (dbl), LSTMTHP10 (dbl), LSTMTHP11 (dbl),
LSTMTHP12
(dbl), USEFSTP3 (dbl), FSTMTHP9 (dbl), FSTMTHP10 (dbl), FSTMTHP11
(dbl),
FSTMTHP12 (dbl), WYNOTUSE (dbl), HPPREGQ (dbl), DURTRY_N (dbl),
DURTRY_P
(dbl), WHYNOUSING1 (dbl), WHYNOUSING2 (dbl), WHYNOUSING3 (dbl),
WHYNOUSING4
(dbl), WHYNOUSING5 (dbl), WHYNOUSING6 (dbl), WHYNOUSING7 (dbl),
WHYNOUSING8
(dbl), MAINNOUSE (dbl), YUSEPILL1 (dbl), YUSEPILL2 (dbl),
YUSEPILL3 (dbl),
YUSEPILL4 (dbl), YUSEPILL5 (dbl), YUSEPILL6 (dbl), IUDTYPE (dbl),
YUSEIUD1
(dbl), YUSEIUD2 (dbl), YUSEIUD3 (dbl), YUSEIUD4 (dbl), YUSEIUD5
(dbl),
EKMETHOD (dbl), CURBCPLC (dbl), NUMPILLS (dbl), CURBCPAY1 (dbl),
CURBCPAY2
(dbl), CURBCPAY3 (dbl), CURBCINS (dbl), NOUSEINS1 (dbl), NOUSEINS2
(dbl),
CURBCAMT (dbl), NOCOST1 (dbl), NOCOST2 (dbl), PST4WKSX (dbl),
PSWKCOND1
(dbl), PSWKCOND2 (dbl), MISSPILL (dbl), P12MOCON (dbl), PXNOFREQ
(dbl),
BTHCON12 (dbl), MEDTST12 (dbl), BCCNS12 (dbl), STEROP12 (dbl),
STCNS12
(dbl), EMCON12 (dbl), ECCNS12 (dbl), PRGTST12 (dbl), PAP12 (dbl),
PELVIC12
(dbl), PRENAT12 (dbl), PARTUM12 (dbl), STDSVC12 (dbl), NUMSVC12
(dbl),
BARRIER_01 (dbl), BARRIER_02 (dbl), BARRIER_03 (dbl), BARRIER_04
(dbl),
BARRIER_05 (dbl), BARRIER_06 (dbl), BARRIER_07 (dbl), BARRIER_08

```

(dbl),
    NUMBCVIS (dbl), BC12PLCX_01 (dbl), BC12PLCX_02 (dbl), BC12PLCX_03
(dbl),
    BC12PLCX_04 (dbl), BC12PLCX_05 (dbl), BC12PLCX_06 (dbl),
BC12PLCX_07 (dbl),
    BC12PLCX_08 (dbl), BC12PLCX_09 (dbl), BC12PLCX_10 (dbl),
BC12PLCX_11 (dbl),
    BC12PLCX_12 (dbl), BC12PLCX_13 (dbl), BC12PLCX_14 (dbl), IDCLINIC
(dbl),
    TALKPROV_1 (dbl), TALKPROV_2 (dbl), TALKPROV_3 (dbl), WHYPSTD
(dbl),
    WHYNOSTD (dbl), BCCLARC (dbl), BC12PAY_01_1 (dbl), BC12PAY_01_2
(dbl),
    BC12PAY_01_3 (dbl), BC12PAY_02_1 (dbl), BC12PAY_02_2 (dbl),
BC12PAY_02_3
    (dbl), BC12PAY_03_1 (dbl), BC12PAY_03_2 (dbl), BC12PAY_03_3 (dbl),
    BC12PAY_04_1 (dbl), BC12PAY_04_2 (dbl), BC12PAY_04_3 (dbl),
BC12PAY_05_1
    (dbl), BC12PAY_05_2 (dbl), BC12PAY_05_3 (dbl), BC12PAY_06_1 (dbl),
    BC12PAY_06_2 (dbl), BC12PAY_06_3 (dbl), BC12PAY_07_1 (dbl),
BC12PAY_07_2
    (dbl), BC12PAY_07_3 (dbl), BC12PAY_08_1 (dbl), BC12PAY_08_2 (dbl),
    BC12PAY_08_3 (dbl), BC12PAY_09_1 (dbl), BC12PAY_09_2 (dbl),
BC12PAY_09_3
    (dbl), BC12PAY_10_1 (dbl), BC12PAY_10_2 (dbl), BC12PAY_10_3 (dbl),
    BC12PAY_11_1 (dbl), BC12PAY_11_2 (dbl), BC12PAY_11_3 (dbl),
BC12PAY_12_1
    (dbl), BC12PAY_12_2 (dbl), BC12PAY_12_3 (dbl), BC12PAY_13_1 (dbl),
    BC12PAY_13_2 (dbl), BC12PAY_13_3 (dbl), BC12PAY_14_1 (dbl),
BC12PAY_14_2
    (dbl), BC12PAY_14_3 (dbl), REGCAR12_F_01 (dbl), REGCAR12_F_02
(dbl),
    REGCAR12_F_03 (dbl), REGCAR12_F_04 (dbl), REGCAR12_F_05 (dbl),
    REGCAR12_F_06 (dbl), REGCAR12_F_07 (dbl), REGCAR12_F_08 (dbl),
    REGCAR12_F_09 (dbl), REGCAR12_F_10 (dbl), REGCAR12_F_11 (dbl),
    REGCAR12_F_12 (dbl), REGCAR12_F_13 (dbl), REGCAR12_F_14 (dbl),
EVERFPC
    (dbl), LASTPAP (dbl), MREASPAP (dbl), AGEFPAP (dbl), AGEFPAP2
(dbl),
    ABNPAP5 (dbl), PELWPAP (dbl), LASTPEL (dbl), MREASPEL (dbl),
AGEFPEL (dbl),
    AGEPEL2 (dbl), EVHPVTST (dbl), HPVWPAP (dbl), LASTHPV (dbl),
MREASHPV
    (dbl), AGEFHPV (dbl), AGEFHPV2 (dbl), SELFTEST (dbl), SELFTEST2
(dbl),
    ASKSMOKE (dbl), ASKPREG (dbl), ASKFOLIC (dbl), TALKDM (dbl),
PROVRESP
    (dbl), PROVSAYBC (dbl), PROVPREBC (dbl), PROVINFOBC (dbl), RWANT
(dbl),

```

PROBWANT (dbl), PWANT (dbl), JINTEND (dbl), JSUREINT (dbl),
JINTENDN (dbl),
JEXPECTL (dbl), JEXPECTS (dbl), JINTNEXT (dbl), INTEND (dbl),
SUREINT
(dbl), INTENDN (dbl), EXPECTL (dbl), EXPECTS (dbl), INTNEXT (dbl),
HLPPRG
(dbl), SEEKCURR (dbl), TYPALLPG_1 (dbl), TYPALLPG_2 (dbl),
TYPALLPG_3
(dbl), TYPALLPG_4 (dbl), TYPALLPG_5 (dbl), TYPALLPG_6 (dbl),
TYPALLPG_7
(dbl), TYPALLPG_8 (dbl), TYPALLPG_9 (dbl), OVUL12M (dbl), INSEM12M
(dbl),
INSCOVPG (dbl), FRSTHELP (dbl), TRYLONG2 (dbl), UNIT_TRYLONG2
(dbl),
HLPPGNOW (dbl), LASTHELP (dbl), HLCMC (dbl), TYPALLMC_1 (dbl),
TYPALLMC_2
(dbl), TYPALLMC_3 (dbl), TYPALLMC_4 (dbl), TYPALLMC_5 (dbl),
MISCNUM (dbl),
INFRTPRB_1 (dbl), INFRTPRB_2 (dbl), INFRTPRB_3 (dbl), INFRTPRB_4
(dbl),
INFRTPRB_5 (dbl), DUCHFREQ (dbl), PID (dbl), PIDSYMPT (dbl), PIDTX
(dbl),
WHENPID (dbl), DIABETES (dbl), GESTDIAB (dbl), UF (dbl), UFSONO
(dbl),
UFCURR (dbl), UFDIAGNOS (dbl), UFLIMIT (dbl), UFTREAT_1 (dbl),
UFTREAT_2
(dbl), UFTREAT_3 (dbl), UFTREAT_4 (dbl), UFTREAT_5 (dbl),
UFTREAT_6 (dbl),
END0 (dbl), ENDOCURR (dbl), ENDODIAG (dbl), ENDOLIM (dbl),
ENDOTREAT_1
(dbl), ENDOTREAT_2 (dbl), ENDOTREAT_3 (dbl), ENDOTREAT_4 (dbl),
ENDOTREAT_5
(dbl), ENDOTREAT_6 (dbl), ENDOTREAT_7 (dbl), ENDOTREAT_8 (dbl),
OVUPROB
(dbl), PCOS (dbl), PCOSSYMP_1 (dbl), PCOSSYMP_2 (dbl), PCOSSYMP_3
(dbl),
PCOSSYMP_4 (dbl), PCOSSYMP_5 (dbl), PCOSSYMP_6 (dbl), PCOSSYMP_7
(dbl),
MENSREG (dbl), MENSEXP_1 (dbl), MENSEXP_2 (dbl), MENSEXP_3 (dbl),
MENSPAIN
(dbl), MENSBCM (dbl), MENSBCAGE (dbl), MENSBCHELP (dbl),
MENSPAIN (dbl),
AGEPELPAIN (dbl), VISION (dbl), HEARING (dbl), MOBILITY (dbl),
COGNITION
(dbl), SELFCARE (dbl), COMMUNIC (dbl), EVRCANCER (dbl), AGECANCER
(dbl),
PRECANCER (dbl), MAMMOG (dbl), MAMMOGN (dbl), AGEMAMMR (dbl),
REASMAMMR
(dbl), FAMHYST (dbl), FAMHYSTA_1 (dbl), FAMHYSTA_2 (dbl),

FAMHYSTA_3 (dbl),
FAMHYSTA_4 (dbl), FAMHYSTA_5 (dbl), MOMRISK70 (dbl), MOMRISK40
(dbl),
ALCORISK (dbl), BCANCRISK (dbl), DONBLOOD (dbl), DONBLDYR (dbl),
HIVTEST
(dbl), WHNHIVTST (dbl), PLCHIV (dbl), RHHIVT1 (dbl), RHHIVT2_1
(dbl),
RHHIVT2_2 (dbl), RHHIVT2_3 (dbl), HIVTST (dbl), PREPHIV (dbl),
PREP12
(dbl), TALKDOCT (dbl), AIDSTALK_01 (dbl), AIDSTALK_02 (dbl),
AIDSTALK_03
(dbl), AIDSTALK_04 (dbl), AIDSTALK_05 (dbl), AIDSTALK_06 (dbl),
AIDSTALK_07
(dbl), AIDSTALK_08 (dbl), AIDSTALK_09 (dbl), AIDSTALK_10 (dbl),
AIDSTALK_11
(dbl), AIDSTALK_12 (dbl), PREGHIV (dbl), EVERVACC (dbl), HPVSHOT1
(dbl),
BLDPRESS (dbl), HIGHBP (dbl), BPMEDS (dbl), BPMON (dbl), BPMONFRQ
(dbl),
USUALCAR (dbl), USLPLACE (dbl), USL12MOS (dbl), CURRCOV (dbl),
COVERHOW_01
(dbl), COVERHOW_02 (dbl), COVERHOW_03 (dbl), COVERHOW_04 (dbl),
PARINSUR
(dbl), COVER12 (dbl), NUMNOCOV (dbl), SAMEADD (dbl), BRNOUT (dbl),
ATTND14
(dbl), FUNDAM_1 (dbl), FUNDAM_2 (dbl), FUNDAM_3 (dbl), FUNDAM_4
(dbl),
RELDLIFE (dbl), ATTNDNOW (dbl), MILSVC (dbl), WRK12MOS (dbl),
FPT12MOS
(dbl), DOLASTWK1 (dbl), DOLASTWK2 (dbl), DOLASTWK3 (dbl),
DOLASTWK4 (dbl),
RWRKST (dbl), RFTPTX (dbl), SPLSTWK1 (dbl), SPLSTWK2 (dbl),
SPLSTWK3 (dbl),
SPLSTWK4 (dbl), SPWRKST (dbl), SPFTPTX (dbl), REACTSLF (dbl),
CHBOTHER
(dbl), CASILANG (dbl), GENHEALT (dbl), BMICAT (dbl), DRWEIGH
(dbl),
TELLWGHT (dbl), WGTSCRN (dbl), ENGSPEAK (dbl), NOBEDYR (dbl),
STAYREL
(dbl), EVSUSPEN (dbl), GRADSUSP (dbl), SMK100 (dbl), AGESMK (dbl),
SMOKE30
(dbl), DRINK12 (dbl), BINGE12 (dbl), POT12 (dbl), VAGSEX (dbl),
AGEVAGR
(dbl), CONDVAG (dbl), WHYCONDL (dbl), GETORALM (dbl), GIVORALM
(dbl),
CONDFELL (dbl), ANYORAL (dbl), J_HADSEX (dbl), TIMING (dbl),
ANALSEX (dbl),
CONDANAL (dbl), OPPSEXANY (dbl), OPPSEXGEN (dbl), CONDSEXL (dbl),
OPPLIFENUM (dbl), OPPYEARNUM (dbl), VAGNUM12 (dbl), ORALNUM12

(dbl),
ANALNUM12 (dbl), RNONMONOG (dbl), BISEXPRT (dbl), NONMONOG (dbl),
NNONMONOG
(dbl), MALSHT12 (dbl), PROSTFRQ (dbl), JOHNFREQ (dbl), GIVORALF
(dbl),
GETORALF (dbl), FEMSEX (dbl), SAMESEXANY (dbl), SAMLIFENUM (dbl),
SAMYEARNUM (dbl), SAMESEX1 (dbl), FSAMEREL (dbl), FEMLEGSTAT
(dbl),
FEMMARRN (dbl), FEMCOHN (dbl), MFLASTP (dbl), DATEAPP (dbl),
ATTRACT (dbl),
ORIENT (dbl), CONFCONC (dbl), TIMALON (dbl), RISKCHEK1 (dbl),
RISKCHEK2
(dbl), RISKCHEK3 (dbl), RISKCHEK4 (dbl), CHLAMTST (dbl), STD0THR12
(dbl),
STDTRT12 (dbl), GON (dbl), CHLAM (dbl), HERPES (dbl), GENWARTS
(dbl),
ABNHPV (dbl), SYPHILIS (dbl), EMOTABUSE (dbl), PHYSABUSE (dbl),
SEXABUSE
(dbl), REVPHYSNEG (dbl), REVEMOTNEG (dbl), WITNESSIPV (dbl),
LIVDRUGS
(dbl), LIVDEPRESS (dbl), SEPJAIL (dbl), RACEDESCRIM (dbl),
GENDDDESCRIM
(dbl), WITVIOL (dbl), EARNTYPE (dbl), EARN (dbl), EARNDK1 (dbl),
EARNDK2
(dbl), EARNDK3 (dbl), EARNDK4 (dbl), TOINCWMY (dbl), TOTINC (dbl),
FMINCDK1
(dbl), FMINCDK2 (dbl), FMINCDK3 (dbl), FMINCDK4 (dbl), FMINCDK5
(dbl),
PUBASST (dbl), FOODSTMP (dbl), WIC (dbl), HLPTRANS (dbl), HLPCHLDC
(dbl),
HLPJOB (dbl), FREEFOOD (dbl), HUNGRY (dbl), MED_COST (dbl),
COVIDVAX (dbl),
COVVAX_M (dbl), COVVAX_Y (dbl), CMCOVVAX (dbl), HADCOVID (dbl),
AGER (dbl),
FMARITAL (dbl), RMARITAL (dbl), HIEDUC (dbl), HIEDUC_I (dbl),
HISPANIC
(dbl), HISPANIC_I (dbl), HISPRACE2 (dbl), HISPRACE2_I (dbl),
NUMKDHH (dbl),
NUMFMHH (dbl), HHFAMTYP (dbl), HHPARTYP (dbl), NCHILDHH (dbl),
HHKIDTYP
(dbl), CSPBBHH (dbl), CSPBSHH (dbl), CSPOKDHH (dbl), INTCTFAM
(dbl),
INTCTFAM_I (dbl), PARAGE14 (dbl), PARAGE14_I (dbl), EDUCMOM (dbl),
EDUCMOM_I (dbl), AGEMOMB1 (dbl), AGEMOMB1_I (dbl), PREGNUM (dbl),
PREGNUM_I
(dbl), COMPREG (dbl), LBPREGS (dbl), LOSSNUM (dbl), ABORTION
(dbl), PARITY
(dbl), BIRTHS5 (dbl), OUTCOM1 (dbl), OUTCOM2 (dbl), OUTCOM3 (dbl),
OUTCOM4

(dbl), OUTCOM5 (dbl), OUTCOM6 (dbl), OUTCOM7 (dbl), OUTCOM8 (dbl),
 OUTCOM9
 (dbl), OUTCOM10 (dbl), OUTCOM11 (dbl), OUTCOM12 (dbl), OUTCOM13
 (dbl),
 OUTCOM14 (dbl), OUTCOM15 (dbl), OUTCOM16 (dbl), DATEND1 (dbl),
 DATEND2
 (dbl), DATEND3 (dbl), DATEND4 (dbl), DATEND5 (dbl), DATEND6 (dbl),
 DATEND7
 (dbl), DATEND8 (dbl), DATEND9 (dbl), DATEND10 (dbl), DATEND11
 (dbl),
 DATEND12 (dbl), DATEND13 (dbl), DATEND14 (dbl), DATEND15 (dbl),
 DATEND16
 (dbl), AGEPRG1 (dbl), AGEPRG2 (dbl), AGEPRG3 (dbl), AGEPRG4 (dbl),
 AGEPRG5
 (dbl), AGEPRG6 (dbl), AGEPRG7 (dbl), AGEPRG8 (dbl), AGEPRG9 (dbl),
 AGEPRG10
 (dbl), AGEPRG11 (dbl), AGEPRG12 (dbl), AGEPRG13 (dbl), AGEPRG14
 (dbl),
 AGEPRG15 (dbl), AGEPRG16 (dbl), DATCON1 (dbl), DATCON2 (dbl),
 DATCON3
 (dbl), DATCON4 (dbl), DATCON5 (dbl), DATCON6 (dbl), DATCON7 (dbl),
 DATCON8
 (dbl), DATCON9 (dbl), DATCON10 (dbl), DATCON11 (dbl), DATCON12
 (dbl),
 DATCON13 (dbl), DATCON14 (lgl), DATCON15 (lgl), DATCON16 (lgl),
 AGECON1
 (dbl), AGECON2 (dbl), AGECON3 (dbl), AGECON4 (dbl), AGECON5 (dbl),
 AGECON6
 (dbl), AGECON7 (dbl), AGECON8 (dbl), AGECON9 (dbl), AGECON10
 (dbl),
 AGECON11 (dbl), AGECON12 (dbl), AGECON13 (dbl), AGECON14 (lgl),
 AGECON15
 (lgl), AGECON16 (lgl), CEBOW (dbl), CEBOWC (dbl), DATBABY1 (dbl),
 AGEBABY1
 (dbl), FMARNO (dbl), CSPBIOKD (dbl), CSPBIOKD_I (dbl), MARDAT01
 (dbl),
 MARDAT01_I (dbl), MARDAT02 (dbl), MARDAT02_I (dbl), MAREND01
 (dbl),
 MAREND01_I (dbl), MAREND02 (dbl), MAREND02_I (dbl), MARDIS01
 (dbl),
 MARDIS01_I (dbl), MARDIS02 (dbl), MARDIS02_I (dbl), FMAR1AGE
 (dbl),
 FMAR1AGE_I (dbl), FMAR2AGE (dbl), AGEDISS1 (dbl), AGEDISS1_I
 (dbl), AGEDD1
 (dbl), AGEDD1_I (dbl), MAR1DISS (dbl), MAR1BIR1 (dbl), MAR1CON1
 (dbl),
 CON1MAR1 (dbl), B1PREMAR (dbl), COHAB1 (dbl), COHAB1_I (dbl),
 COHSTAT
 (dbl), COHSTAT_I (dbl), COHOUT (dbl), COH1DUR (dbl), COH1DUR_I

```

(dbl),
    HADSEX (dbl), SEXONCE (dbl), SEXONCE_I (dbl), VRY1STSX (dbl),
VRY1STSX_I
    (dbl), VRY1STAG (dbl), SEXMAR (dbl), SEXUNION (dbl), FPDUR (dbl),
FPDUR_I
    (dbl), LSEXDATE (dbl), LSEXDATE_I (dbl), LSEXRAGE (dbl), LSEXPREG
(dbl),
    SEX3MO (dbl), PARTS1YR (dbl), PARTS1YR_I (dbl), PARTDUR1 (dbl),
PARTDUR2
    (dbl), PARTDUR3 (dbl), PARTDUR1_I (dbl), PARTDUR2_I (dbl),
PARTDUR3_I
    (dbl), RELATP1 (dbl), RELATP2 (dbl), RELATP3 (dbl), RELATP1_I
(dbl),
    RELATP2_I (dbl), RELATP3_I (dbl), LIFPRTNR (dbl), LIFPRTNR_I
(dbl),
    NUMP3MOS (dbl), NUMP3MOS_I (dbl), STRLOPER (dbl), ANYMTHD (dbl),
ANYMTHD_I
    (dbl), FECUND (dbl), FECUND_I (dbl), ANYBC36 (dbl), CMSTRTHP
(dbl), NOSEX36
    (dbl), NOSEX12 (dbl), NOSEX12_I (dbl), INFERT (dbl), INFERT_I
(dbl),
    ANYBC12 (dbl), SEXP3MO (dbl), SEXP3MO_I (dbl), CONSTAT1 (dbl),
CONSTAT2
    (dbl), CONSTAT3 (dbl), CONSTAT4 (dbl), CONSTAT1_I (dbl),
CONSTAT2_I (dbl),
    CONSTAT3_I (dbl), CONSTAT4_I (dbl), SEX1MTHD1 (dbl), SEX1MTHD2
(dbl),
    SEX1MTHD3 (dbl), SEX1MTHD4 (dbl), SEX1MTHD1_I (dbl), SEX1MTHD2_I
(dbl),
    SEX1MTHD3_I (dbl), SEX1MTHD4_I (dbl), METH12M1 (dbl), METH12M2
(dbl),
    METH12M3 (dbl), METH12M4 (dbl), METH12M1_I (dbl), METH12M2_I
(dbl),
    METH12M3_I (dbl), METH12M4_I (dbl), METH3M1 (dbl), METH3M2 (dbl),
METH3M3
    (dbl), METH3M4 (dbl), METH3M1_I (dbl), METH3M2_I (dbl), METH3M3_I
(dbl),
    METH3M4_I (dbl), DATEUSE1 (dbl), DATEUSE1_I (dbl), MAROUT1 (dbl),
MAROUT2
    (dbl), MAROUT3 (dbl), MAROUT4 (dbl), MAROUT5 (dbl), MAROUT6 (dbl),
MAROUT7
    (dbl), MAROUT8 (dbl), MAROUT9 (dbl), MAROUT10 (dbl), MAROUT11
(dbl),
    MAROUT12 (dbl), MAROUT13 (dbl), MAROUT14 (lgl), MAROUT15 (lgl),
MAROUT16
    (lgl), MARCON1 (dbl), MARCON2 (dbl), MARCON3 (dbl), MARCON4 (dbl),
MARCON5
    (dbl), MARCON6 (dbl), MARCON7 (dbl), MARCON8 (dbl), MARCON9 (dbl),
MARCON10

```

(dbl), MARCON11 (dbl), MARCON12 (dbl), MARCON13 (dbl), MARCON14
 (lgl),
 MARCON15 (lgl), MARCON16 (lgl), NWWANTRP1 (dbl), NWWANTRP2 (dbl),
 NWWANTRP3
 (dbl), NWWANTRP4 (dbl), NWWANTRP5 (dbl), NWWANTRP6 (dbl),
 NWWANTRP7 (dbl),
 NWWANTRP8 (dbl), NWWANTRP9 (dbl), NWWANTRP10 (dbl), NWWANTRP11
 (dbl),
 NWWANTRP12 (dbl), NWWANTRP13 (dbl), NWWANTRP14 (lgl), NWWANTRP15
 (lgl),
 NWWANTRP16 (lgl), WANTRP1 (dbl), WANTRP2 (dbl), WANTRP3 (dbl),
 WANTRP4
 (dbl), WANTRP5 (dbl), WANTRP6 (dbl), WANTRP7 (dbl), WANTRP8 (dbl),
 WANTRP9
 (dbl), WANTRP10 (dbl), WANTRP11 (dbl), WANTRP12 (dbl), WANTRP13
 (dbl),
 WANTRP14 (lgl), WANTRP15 (lgl), WANTRP16 (lgl), WANTP1 (dbl),
 WANTP2 (dbl),
 WANTP3 (dbl), WANTP4 (dbl), WANTP5 (dbl), WANTP6 (dbl), WANTP7
 (dbl),
 WANTP8 (dbl), WANTP9 (dbl), WANTP10 (dbl), WANTP11 (dbl), WANTP12
 (dbl),
 WANTP13 (dbl), WANTP14 (lgl), WANTP15 (lgl), WANTP16 (lgl),
 FPTITBC (dbl),
 FPTITBC_I (dbl), FPTITCHK (dbl), FPTITCHK_I (dbl), FPTITCBC (dbl),
 FPTITCBC_I (dbl), FPTITSTE (dbl), FPTITSTE_I (dbl), FPTITCST
 (dbl),
 FPTITCST_I (dbl), FPTITEC (dbl), FPTITEC_I (dbl), FPTITCEC (dbl),
 FPTITCEC_I (dbl), FPTITPRE (dbl), FPTITPRE_I (dbl), FPTITPAP
 (dbl),
 FPTITPAP_I (dbl), FPTITPEL (dbl), FPTITPEL_I (dbl), FPTITPRN
 (dbl),
 FPTITPRN_I (dbl), FPTITPPR (dbl), FPTITPPR_I (dbl), FPTITSTD
 (dbl),
 FPTITSTD_I (dbl), INTENT (dbl), ADDEXP (dbl), ADDEXP_I (dbl),
 EVHIVTST
 (dbl), FPTITHIV (dbl), FPTITHIV_I (dbl), CURR_INS (dbl),
 CURR_INS_I (dbl),
 RELIGION (dbl), RELIGION_I (dbl), LABORFOR (dbl), LABORFOR_I
 (dbl), TOTINCR
 (dbl), TOTINCR_I (dbl), POVERTY (dbl), POVERTY_I (dbl), METRO
 (dbl),
 WGT2022_2023 (dbl), VEST (dbl), VECL (dbl), CMINTVW (dbl), CMLSTYR
 (dbl),
 CMJAN3YR (dbl), CMJAN4YR (dbl), CMJAN5YR (dbl), CMFIVYR (dbl),
 YEAR (dbl),
 QUARTER (dbl), PHASE1 (dbl), PHASE2 (dbl), PHASE3 (dbl),
 contraception_group (chr), marital_fact (fct), health_ord (ord),
 race_fact


```

(fct)
              Coef      SE      z      p
(Intercept):1 -1.5673702 0.3157962 -4.9632 6.933e-07
(Intercept):2 -1.2079002 0.1540699 -7.8399 4.507e-15
(Intercept):3 -5.9694711 0.2731543 -21.8538 < 2.2e-16
AGER:1        -0.0516788 0.0133497 -3.8712 0.0001083
AGER:2        -0.0088406 0.0059231 -1.4925 0.1355560
AGER:3         0.1440355 0.0069347 20.7703 < 2.2e-16
HIEDUC:1       0.0618331 0.0509379  1.2139 0.2247887
HIEDUC:2       0.1557643 0.0187280  8.3172 < 2.2e-16
HIEDUC:3      -0.0863227 0.0228815 -3.7726 0.0001616
race_factNH White:1 -0.5995477 0.2798893 -2.1421 0.0321863
race_factNH White:2  0.1225205 0.0921489  1.3296 0.1836526
race_factNH White:3  0.5375215 0.1898626  2.8311 0.0046387
race_factNH Black:1 -0.0292481 0.3843816 -0.0761 0.9393464
race_factNH Black:2 -0.3731177 0.1145929 -3.2560 0.0011298
race_factNH Black:3 -0.1289192 0.2047595 -0.6296 0.5289482
race_factOther:1   -0.4947731 0.3971104 -1.2459 0.2127888
race_factOther:2   -0.2536824 0.1037629 -2.4448 0.0144922
race_factOther:3   -0.5395499 0.2980356 -1.8104 0.0702409

```

```

nommod_summary <- cbind(
  "Log-Odds" = coef(nom_mod),
  "Odds Ratio" = exp(coef(nom_mod))
)

```

```

round(nommod_summary, 3)

```

	Log-Odds	Odds Ratio
(Intercept):1	-1.567	0.209
(Intercept):2	-1.208	0.299
(Intercept):3	-5.969	0.003
AGER:1	-0.052	0.950
AGER:2	-0.009	0.991
AGER:3	0.144	1.155
HIEDUC:1	0.062	1.064
HIEDUC:2	0.156	1.169
HIEDUC:3	-0.086	0.917
race_factNH White:1	-0.600	0.549
race_factNH White:2	0.123	1.130
race_factNH White:3	0.538	1.712
race_factNH Black:1	-0.029	0.971
race_factNH Black:2	-0.373	0.689
race_factNH Black:3	-0.129	0.879
race_factOther:1	-0.495	0.610
race_factOther:2	-0.254	0.776
race_factOther:3	-0.540	0.583

Intepretation:

Looking at the Odd ratios for the Ordinal Logistic Regression the AGER variable is at 1.030, showing that for every year increase in age, the chances of being in a poorer health category increases.

For the HIEDUC the Odd ratio is at 0.899, showing that an increase in higher education, the odds of reporting poorer health decreases.

The t-values show that the AGER and the HIEDUC are from zero indicating that these are predictors that are significant contributors to the model's power. The same can be said of the Intercept Thresholds where their t-values show that they are statistically significant.

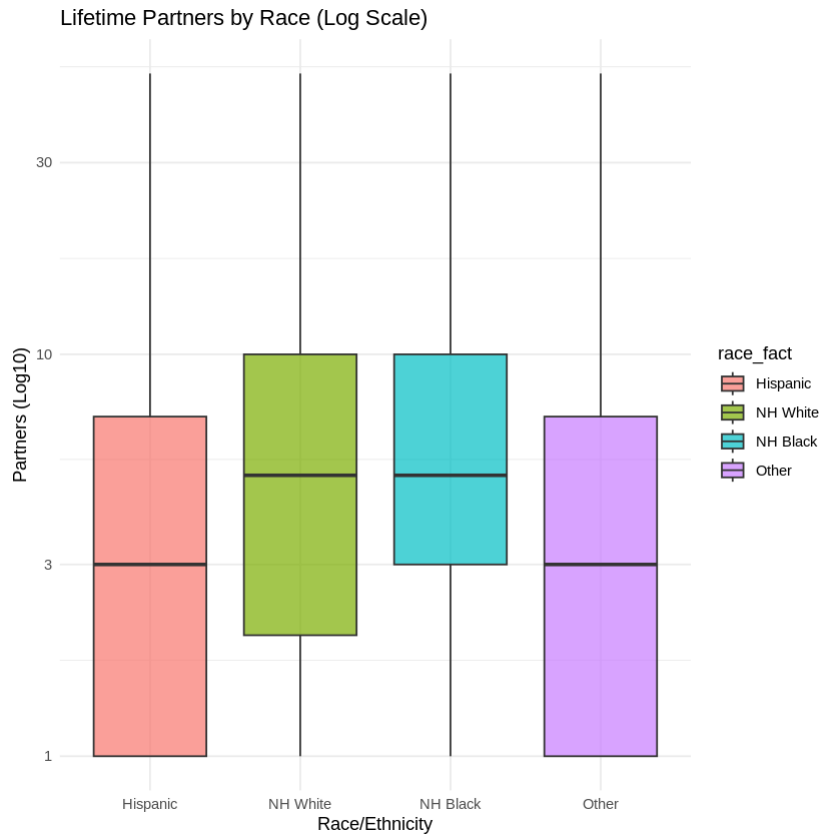
For the Multinomial Logistic Regression, the Odd ratios for AGER shows it is a strong predictor of sterilization with the log-odds of sterilization of Group 1 relative to None/Other decreases as age increases, but the strong positive coefficient for Group 3 suggest that younger groups more on Other/None than barriers compared to the reference

With the HIEDUC the Odd ratio shows that a higher education increases the odds of using Hormonal/LARC methods compared to the Other/None group.

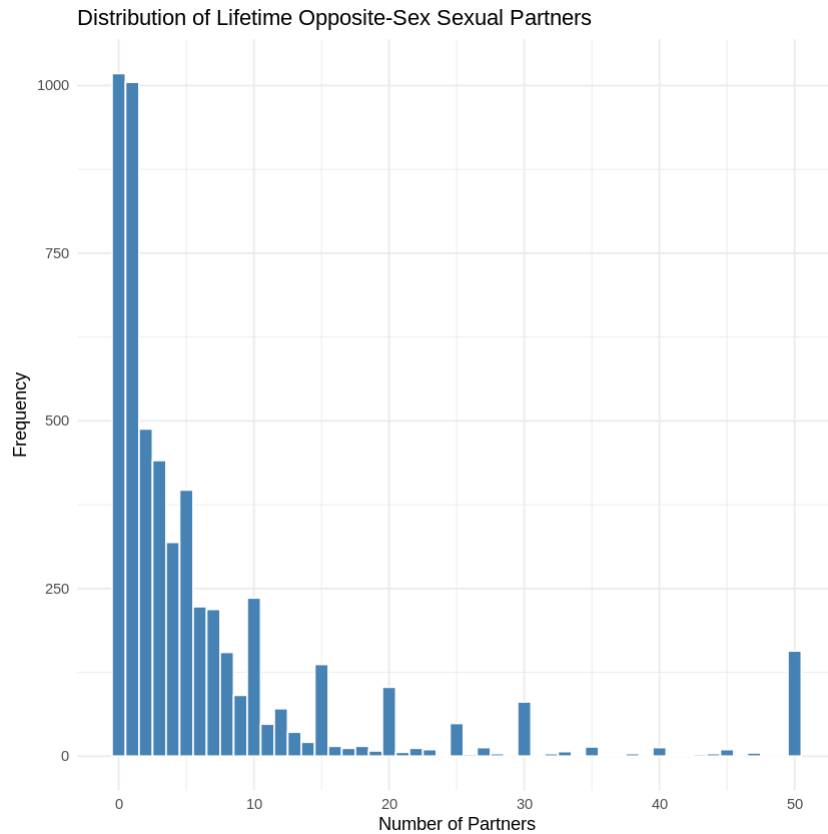
The z-values show that AGER and HIEDUC are essentially 0, meaning that they are improving the fit of the model compared to without them.

D. Exploratory Data Analysis & Multiple Comparisons

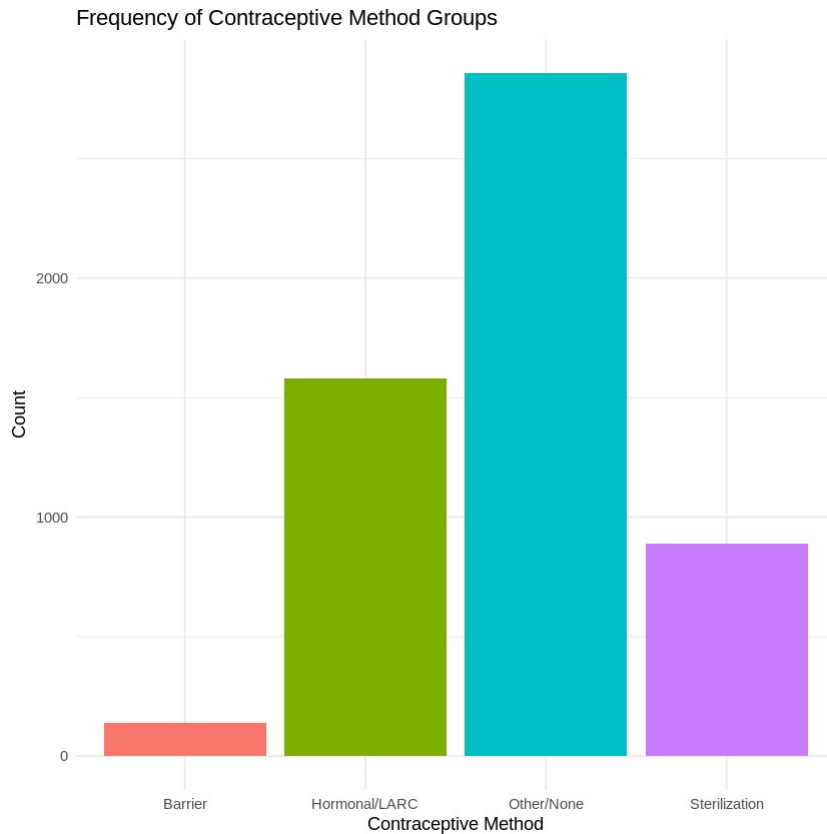
```
df_clean %>%  
  filter(LIFPRTNR > 0) %>%  
  ggplot(aes(x = race_fact, y = LIFPRTNR, fill = race_fact)) +  
  geom_boxplot(alpha = 0.7) +  
  scale_y_log10() +  
  labs(title = "Lifetime Partners by Race (Log Scale)",  
        x = "Race/Ethnicity", y = "Partners (Log10)") +  
  theme_minimal()
```



```
df_clean %>%
  ggplot(aes(x = LIFPRTNR)) +
  geom_histogram(binwidth = 1, fill = "steelblue", color = "white") +
  labs(title = "Distribution of Lifetime Opposite-Sex Sexual
Partners",
       x = "Number of Partners",
       y = "Frequency") +
  theme_minimal()
```



```
df_clean %>%  
  ggplot(aes(x = contraception_group, fill = contraception_group)) +  
  geom_bar() +  
  labs(title = "Frequency of Contraceptive Method Groups",  
        x = "Contraceptive Method",  
        y = "Count") +  
  theme_minimal() +  
  theme(legend.position = "none")
```



```
pairwise.t.test(df_clean$LIFPRTNR, df_clean$race_fact, p.adjust.method = "bonferroni")
```

Pairwise comparisons using t tests with pooled SD

data: df_clean\$LIFPRTNR and df_clean\$race_fact

	Hispanic	NH White	NH Black
NH White	5.3e-10	-	-
NH Black	2.3e-05	1.0000	-
Other	1.0000	7.1e-06	0.0013

P value adjustment method: bonferroni

Discussion:

When doing multiple statistical tests the chances of running into a Type-1 error increases (also known as a "false positive").

This can be alleviated using the bonferroni method which reduces this risk and makes it harder to achieve. Allowing for the tests to remain reliable and not due to any luck or unluck.